

Question 2024(3rd)

Question 1

a) What is a system? Describe the ways to study a system.

(6 marks)

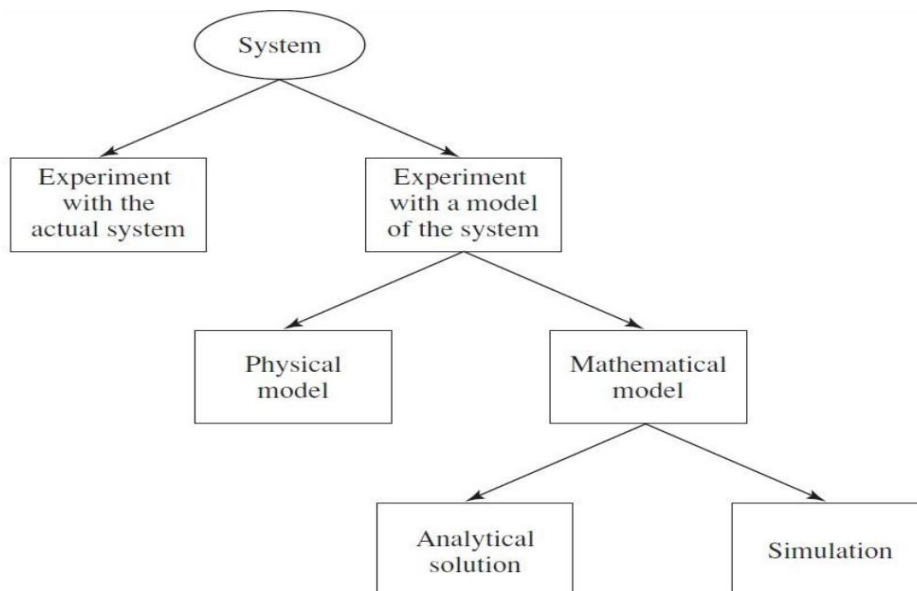
b) What are the desirable features of a simulation software?

(6 marks)

a) A system is defined to be a collection of entities, such as people or machines, that interact together toward the accomplishment of some logical end.

We categorize systems to be of two types, **discrete and continuous**. A discrete system is one for which the state variables change instantaneously at separated points in time. A continuous system is one for which the state variables change continuously with respect to time.

The ways to study a system.



Experiment with the Actual System vs. Experiment with a Model of the System

If it is possible (and cost- effective) to change the system physically and then run it under the new conditions, it is usually better to do that. This is because there is no doubt about whether the results we study are valid.

However, in most cases this is not practical, because such an experiment can be too expensive or can disturb the system too much.

Physical Model vs. Mathematical

The vast majority of models built for such purposes are mathematical, representing a system in terms of logical and quantitative relationships.

That system are then manipulated and changed to see how the model reacts, which helps us understand how the real system would behave.

Analytical Solution vs. Simulation

If the model is simple enough, it may be possible to work with its relationships and quantities to get an exact, mathematical model.

$$-y=mx+c$$

However, many real systems are very complex. Because of this, their mathematical models also become very complex, and it is not possible to find an exact analytical solution. In such cases, the model is studied using simulation. This means the model is tested numerically by giving it different inputs and observing how these inputs affect the system's performance.

b) Desirable Features of a Simulation Software (6 marks)

There are numerous features to consider when selecting simulation software. The desirable features of a simulation software can be grouped as follows:

1. General Capabilities

The most important feature is modeling flexibility, i.e., the ability to model systems of any complexity. The software should allow defining and changing entity attributes and global variables, using decision logic (if-then-else), mathematical expressions, and functions. It should also support hierarchical modeling, fast execution speed, good debugging tools, ease of use with a graphical user interface, and the ability to create user-friendly front ends and run-time versions.

2. Hardware and Software Requirements

The software should be compatible with commonly used computer platforms and operating systems, require reasonable RAM, and preferably support parallel processing on multiple cores or networked computers.

3. Animation and Dynamic Graphics

Built-in animation should be available to visually represent system behavior over time. Animation helps in understanding the model, debugging, training users, validating the model, and communicating results to nontechnical users.

4. Statistical Capabilities

Good statistical analysis features are essential. The software must have a reliable random-number generator and tools to compute correct performance measures and statistical summaries.

5. Customer Support and Documentation

The vendor should provide good documentation, online help, detailed examples, regular training, and quick technical support to assist users in learning and using the software effectively.

6. Output Reports and Graphics

The software should generate standard and customizable reports, along with graphical outputs such as histograms, time plots, bar charts, pie charts, and correlation plots to clearly present simulation results.

Question 2

a) What are the key differences between the event-scheduling approach and the process-interaction approach? In which ways is the process approach considered better?

(8 marks)

b) Differentiate between deterministic and stochastic simulation models.

(4 marks)

a) Key differences between the event-scheduling approach and the process-interaction approach, and why the process approach is considered better (8 marks)

Differences between the event-scheduling approach and the process-interaction approach---

Event-Scheduling Approach

- In the event-scheduling approach, a system is modeled by first identifying its main events.
- For each event, an event routine is written that describes the state changes that occur when the event happens.
- The simulation advances in time by executing events one by one, in increasing order of their scheduled occurrence time.
- A key property of an event routine is that no simulated time passes while it is being executed.
- This approach focuses on what happens at specific event times, rather than on the life of individual entities.

Process-Interaction Approach

- In the process approach, a system is modeled as a collection of processes.
- A process is a time-ordered sequence of interrelated events, separated by intervals of time, describing the complete experience of an entity as it moves through a system.
- For example, the process of a customer arriving, waiting, receiving service, and leaving in a single-server system can be modeled as one process.
- A simulation model may contain several different types of processes.
- For each process, there is a process routine that describes the entire history of the process-entity.
- A process routine clearly shows how simulated time passes and usually has more than one entry point, meaning the process can start or continue from different steps during the simulation.

Why the Process Approach Is Considered Better

- The process approach is more natural and intuitive, because it models the system from the entity's point of view.
- It is easier to understand, develop, and modify, especially for large and complex systems.

- The explicit handling of time delays inside the process routines makes the model easier to understand and read.
- Because of these advantages, most modern simulation packages use the process-interaction approach instead of the event-scheduling approach.

Question 3

a) Why are calibration and validation of models necessary? How are they done in reality?

(8 marks)

b) Which one is more suitable for defense-related simulation: simulation software or programming languages? Argue your answer.

(4 marks)

a)

Calibration means adjusting the simulation model so that its output matches real-world data as closely as possible. Validation means checking whether the simulation model correctly represents the real system.

Why Calibration and Validation are Necessary

- **Avoiding Erroneous Results:** Validation is essential because if a model or the software used is inflexible or inaccurate, the study may produce wrong or erroneous results, leading to poor decision-making.
- **Ensuring Result Validity:** Since many real-life systems involve randomness, the results from a simulation are only estimates, not exact values. Therefore, validation is needed to make sure the simulation results truly represent how the real system behaves.
- **Statistical Dependability:** A simulation is only as good as its underlying statistical components; without a reliable random-number generator, the results are considered completely invalid.
- **Risk Mitigation:** In cases where experimenting with a real system is too expensive or disruptive, such as changing the number of tellers in a bank, validation provides the confidence needed to trust the model's predictions before implementation.

How They are Done in Reality

- **Comparison with Actual Systems:** When possible, the most direct way to validate a model is to run it under known conditions and compare the results to the performance of the actual, physical system.
- **Visual Verification via Animation:** In practice, animation is a vital tool for showing that a model is valid (or invalid) by allowing observers to see if the icons (like forklifts or machines) behave incorrectly compared to reality.
- **Testing Logic and Relationships:** Analysts manipulate the model's logical and quantitative relationships to observe how it reacts, checking if the simulated behaviour matches how they expect the real system to behave.

- **Iterative Numerical Testing:** Calibration is performed by providing the model with different inputs and observing how these affect the system's performance, adjusting parameters until the outputs are valid.

b) Which one is more suitable for defense-related simulation software or programming languages? (4 marks)

For defense-related simulation, a general-purpose programming language such as C, C++, or Java is often more suitable.

This is because most modelers already know programming languages, while they may not be familiar with a simulation package. A simulation model written efficiently in C, C++, or Java usually requires less execution time, which is important for complex and large-scale defense simulations. Programming languages also allow greater flexibility and can be closely tailored to a specific defense application, unlike simulation packages that use one common set of modeling constructs for many systems.

Moreover, C++ and Java are object-oriented, which is very important in the defense industry for modeling complex systems. Since most simulation packages are not truly object-oriented, programming languages are often preferred for defense-related simulations.

Question 4

a) Describe the simulation of a single-server queuing system.
(6 marks)

b) Describe the simulation of an inventory system.
(6 marks)

a) Describe the simulation of a single-server queuing system (8 marks)

A single-server queuing system consists of one server that provides service to arriving customers, such as a one-operator barbershop. Although the system is simple, its simulation method represents how more complex systems are simulated.

In this system, interarrival times $A_1, A_2, \dots$ are independent and identically distributed (IID) random variables, meaning each arrival time gap follows the same probability distribution and does not depend on others. When a customer arrives and finds the server idle, the customer enters service immediately. The service times $S_1, S_2, \dots$ are also IID random variables and are independent of the interarrival times.

If a customer arrives while the server is busy, the customer joins the end of a single queue. When the server finishes serving a customer, it selects the next customer from the queue using the first-in, first-out (FIFO) rule. This ensures customers are served in the order they arrive.

The simulation begins in an empty-and-idle state, meaning there are no customers in the system and the server is idle. At time 0, the simulation waits for the first customer to arrive after the first interarrival time A_1 , rather than assuming an arrival exactly at time 0.

The system is simulated until a fixed number n of customers have completed their delays in queue. The time at which the simulation ends is a random variable, since it depends on the randomly

generated interarrival and service times. During the simulation, performance measures such as average delay in queue and average number of customers in the queue can be estimated to evaluate system behavior.

This step-by-step event-based approach clearly demonstrates how a single-server queuing system is simulated and analyzed.

b) Describe the simulation of an inventory system.

The simulation of an inventory system is used to analyze and compare different inventory ordering policies. In this system, customer demands occur randomly over time, where the time between successive demands follows an exponential distribution. The demand size is also random and can take different values with given probabilities.

At the beginning of each month, the inventory level is reviewed and the company follows a stationary (s, S) inventory policy to determine the order quantity. Specifically,

$$Z = \begin{cases} S - I, & \text{if } I < s \\ 0, & \text{if } I \geq s \end{cases}$$

Here, I represents the inventory level at the beginning of the month. The parameter s (small s) is called the reorder point, which is the minimum inventory level. The parameter S (capital S) is known as the order-up-to level, which represents the target inventory level after ordering. If the inventory level I is less than the reorder point s , an order of size $S - I$ is placed; otherwise, no order is placed.

Each order incurs a fixed setup cost and a variable cost per item, and the delivery arrives after a random lead time.

When a demand occurs, it is immediately satisfied if enough inventory is available. If inventory is insufficient, the unmet demand is backlogged, leading to a negative inventory level. When an order arrives, the backlog is cleared first and any remaining items are added to inventory.

The model also includes holding costs for positive inventory and shortage costs for backlogged items.

By simulating the system over a fixed period, the average ordering, holding, and shortage costs are calculated to evaluate the performance of different inventory policies.

Question 5

a) In a single-server queuing system, arrivals occur at times 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, and 7.2.

Departures (service completions) occur at times 2.4, 3.1, 3.3, 5.6, and 9.0.

If the simulation ends at time $T(n) = 9.0$, compute $\hat{q}(n)$.
(6 marks)

b) A service facility consists of two servers in series (tandem), each with its own FIFO queue. A customer completing service at server 1 proceeds to server 2, while a customer completing service at server 2 leaves the facility.

Assume:

- Interarrival times to server 1 are IID exponential random variables with mean 1 minute.
- Service times at server 1 are IID exponential random variables with mean 0.5 minute.
- Service times at server 2 are IID exponential random variables with mean 1.0 minute.

For x minutes, estimate for **each server**:

1. The expected average delay in queue of a customer
2. The expected time-average number of customers in queue
3. The expected utilization

(6 marks)

a) Computation of $\hat{q}(n)$

We are asked to compute the **time-average number of customers in the queue**, $\hat{q}(n)$, for a **single-server queuing system** over the simulation period

$$T(n) = 9.0$$

The formula used is:

$$\hat{q}(n) = \frac{1}{T(n)} \int_0^{T(n)} Q(t) dt$$

That is, **area under the queue-length curve divided by total simulation time**.

Step 1: Track Queue Length Over Time

Time Interval	$Q(t)$	Duration	Area
0.0 – 0.4	0	0.4	0
0.4 – 1.6	0	1.2	0
1.6 – 2.1	1	0.5	0.5
2.1 – 2.4	2	0.3	0.6
2.4 – 3.1	1	0.7	0.7
3.1 – 3.3	0	0.2	0
3.3 – 3.8	0	0.5	0
3.8 – 4.0	0	0.2	0
4.0 – 5.6	1	1.6	1.6
5.6 – 5.8	1	0.2	0.2
5.8 – 7.2	2	1.4	2.8
7.2 – 9.0	3	1.8	5.4

Total area =

$$0.5 + 0.6 + 0.7 + 1.6 + 0.2 + 2.8 + 5.4 = 11.8$$

Step 3: Compute $\hat{q}(n)$

$$\hat{q}(n) = \frac{11.8}{9.0} = 1.31$$

Final Answer (6 Marks)

$$\boxed{\hat{q}(n) = 1.31}$$

(All other intervals have queue length 0.)

Interpretation:

On average, about **1.31 customers** were waiting in the queue during the simulation period.

b) Tandem Queue System

System Description

The system consists of **two servers in series (tandem)**, each with its own **FIFO queue**.

Flow of customers:

External arrivals $\rightarrow$ Queue 1 $\rightarrow$ Server 1 $\rightarrow$ Queue 2 $\rightarrow$ Server 2 $\rightarrow$ Departure

Given Parameters

- **Interarrival times to Server 1:** Exponential with mean **1 minute**

$$\lambda = 1 \text{ customer/minute}$$

- **Service times at Server 1:** Exponential with mean **0.5 minute**

$$\mu_1 = \frac{1}{0.5} = 2 \text{ customers/minute}$$

- **Service times at Server 2:** Exponential with mean **1.0 minute**

$$\mu_2 = \frac{1}{1.0} = 1 \text{ customer/minute}$$

1. Stability Check and Jackson Network

This system is an **open Jackson network** with:

- Poisson external arrivals,
- Exponential service times,
- Deterministic routing from Server 1 to Server 2.

Effective arrival rates

$$r_1 = \lambda = 1, r_2 = r_1 = 1$$

Traffic intensities

$$\rho_1 = \frac{r_1}{\mu_1} = \frac{1}{2} = 0.5 < 1 (\text{stable})$$
$$\rho_2 = \frac{r_2}{\mu_2} = \frac{1}{1} = 1 (\text{unstable})$$

Since $\rho_2 = 1$, a **steady-state does not exist** for Server 2 in an M/M/1 queue with infinite buffer, and the queue length grows without bound.

2. Expected Performance Measures (M/M/1 Formulas)

For an M/M/1 queue:

$$L_q = \frac{\rho^2}{1 - \rho}, W_q = \frac{L_q}{r}, \text{Utilization} = \rho$$

Server 1 (Stable)

$$\rho_1 = 0.5$$
$$L_{q1} = \frac{(0.5)^2}{1 - 0.5} = \frac{0.25}{0.5} = 0.5 \text{ customers}$$
$$W_{q1} = \frac{L_{q1}}{r_1} = \frac{0.5}{1} = 0.5 \text{ minutes}$$
$$\text{Utilization}_1 = 0.5$$

Server 2 (Unstable, $\rho_2 = 1$)

- Expected queue length:

$$L_{q2} \rightarrow \infty$$

- Expected waiting time in queue:

$$W_{q2} \rightarrow \infty$$

- Utilization:

$$\text{Utilization}_2 = 1$$

3. Summary of Results

Server	Expected avg. delay in queue W_q	Expected time-avg. number in queue L_q	Expected utilization ρ
1	0.5 minutes	0.5 customers	0.5 (50%)
2	Infinite	Infinite	1.0 (100%)

Question 6

- a) Write down the properties of random numbers.
(4 marks)

b) Discuss the techniques for generating random numbers with suitable examples.
(8 marks)

a) Properties of Random Numbers (4 Marks)

In simulation, the quality of random numbers depends on the properties of the random-number generator that produces them. A good arithmetic random-number generator must satisfy the following essential properties to ensure valid and reliable simulation results:

- **Statistical Quality:** The generated numbers should appear to be uniformly distributed on the interval (0,1) and should not be correlated with one another. If this condition is violated, simulation results may become completely unreliable.
- **Efficiency:** The generator should be computationally fast and require minimal computer storage, making it practical for large simulation studies.
- **Reproducibility:** The generator must be able to reproduce the exact same sequence of random numbers. This is crucial for debugging simulation programs and for making fair and accurate comparisons between alternative system designs using identical random inputs.
- **Portability and Streams:** The generator should be portable, producing the same results on different computers and compilers. It should also support multiple independent streams, allowing separate random-number sequences to be used for different components of the simulation model.

b) Techniques for Generating Random Numbers (8 Marks)

Random numbers U_i are usually generated using arithmetic methods. These methods use fixed mathematical formulas to create a sequence of numbers. Although the process is deterministic (not truly random), the generated numbers look random and behave like independent values uniformly distributed between 0 and 1.

1. Linear Congruential Generators (LCGs):

The most widely used technique, where a sequence of integers Z_i is generated using

$$Z_i = (aZ_{i-1} + c) \pmod{m}$$

Here, m is the modulus, a the multiplier, c the increment, and Z_0 the seed. The random number is then obtained as $U_i = Z_i/m$.

Example:

For $m = 16$, $a = 5$, $c = 3$, and $Z_0 = 7$:

$$Z_1 = (5 \times 7 + 3) \pmod{16} = 6$$

Hence, $U_1 = 6/16 = 0.375$.

2. Mixed and Multiplicative Generators:

If $c > 0$, the generator is called mixed and may achieve a full period of length m .

If $c = 0$, it is a multiplicative generator, which is more computationally efficient but cannot achieve a full period.

3. Multiple Recursive Generators (MRGs)

To obtain longer periods than standard LCGs, Multiple Recursive Generators use a linear combination of several previous values instead of only the last one.

- **Formula:**

$$Z_i = (a_1 Z_{i-1} + a_2 Z_{i-2} + \dots + a_q Z_{i-q}) \bmod m$$

- With suitable choices of parameters, the period can be as large as $m^q - 1$. MRGs are preferred because they are portable across systems and perform well in statistical tests.

4. Composite Generators:

These combine two or more generators to improve statistical quality and extend the period.

- **Shuffling:** One generator is used to randomly shuffle the outputs of another generator.
- **Combined MRGs:** This approach combines J different MRGs to produce extremely long periods.

For example, a generator combining two MRGs ($J = 2$) can achieve a period of approximately 2^{191} and shows excellent statistical performance.

5. Feedback Shift Register Generators:

These are another class of generators mentioned as alternatives for producing random sequences in simulation modeling.

Conditions for a good LCG (exam-important)

For

$$Z_i = (aZ_{i-1} + c) \bmod m$$

1. c and m must be relatively prime

$$\gcd(c, m) = 1$$

2. $(a-1)$ must be divisible by all prime factors of m
3. If m is divisible by 4, then $(a-1)$ must also be divisible by 4

If these are satisfied $\rightarrow$ maximum period = m

Question 7

- a) What properties should a “good” arithmetic random-number generator possess? (4 marks)
 - b) What is the concept of composite generators? Describe an existing one. (8 marks)
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a) Properties of a “good” arithmetic random-number generator (5 marks)

A “good” arithmetic random-number generator should have the following properties:

1. The generated numbers should appear to be uniformly distributed on $[0, 1]$ and should not be correlated with each other; otherwise, the simulation results may be completely invalid.
2. From a practical point of view, the generator should be fast and should not require a large amount of storage.
3. It should be possible to reproduce the same sequence of random numbers exactly, which helps in debugging and verifying programs and allows fair and accurate comparison of different systems using identical random numbers.
4. The generator should allow the user to easily create separate “streams” of random numbers, which can be treated as independent generators.
5. The generator should be portable, meaning it should produce the same sequence of random numbers on different computers and with different standard compilers.

b) Concept of composite generators and description of an existing one (8 marks)

Concept of composite generators

A composite generator is a random-number generator that combines two or more separate generators to produce the final random numbers. The main goal of using a composite generator is to obtain a much longer period and better statistical properties than those achieved by any single simple generator alone.

The main disadvantage of composite generators is that the computational cost of generating each random number U_i is usually higher than when using only one simple generator.

Example of an Existing Composite Generator

1. Shuffled Linear Congruential Generator

One of the earliest composite generators uses a shuffling technique:

- First, a vector

$$V = (V_1, V_2, \dots, V_k)$$

is filled with the first k random numbers produced by a first LCG (usually $k = 128$).

- A second LCG is then used to generate a random index i , where

$$i \in \{1, 2, \dots, k\}.$$

- The value V_i is used to generate next $U(0,1)$ random number.
- The first LCG generates a new random number and replaces the value at position i in the vector V .
- This process is repeated for each new random number.

This shuffling method helps to reduce correlations and increase the period of the generator. However, shuffled generators are harder to analyze, and unlike simple LCGs, it is not possible to jump directly to a future value without generating all previous values.

2. Combined Multiple Recursive Generator (MRG)

Another important composite generator is the combined multiple recursive generator (MRG) proposed by L'Ecuyer (1999). In this method, two MRGs ($J = 2$) are combined in a special way. When the parameters are chosen carefully, this generator achieves an extremely long period of about 2^{191} and shows excellent statistical properties. Because of this, it is one of the most successful and widely used composite generators in practice.

Two separate multiple recursive generators produce integer sequences $\{Z_{1,i}\}$ and $\{Z_{2,i}\}$, each using its own recurrence relation and modulus.

At each step, the two sequences are combined using subtraction modulo a large number:

$$Y_i = (Z_{1,i} - Z_{2,i}) \bmod m$$

The final uniform random number is obtained by normalization:

$$U_i = \frac{Y_i}{m}$$

Question 8

a) Suppose the five numbers 0.45, 0.80, 0.15, 0.05, and 0.95 were generated. Perform a test for uniformity using the Kolmogorov–Smirnov test with a level of significance $\alpha = 0.05$.

(6 marks)

b) Show and briefly discuss model building, verification, and validation.

(6 marks)

b) Model Building, Verification, and Validation (06)

In simulation studies, model building, verification, and validation are closely related steps that are repeatedly performed to ensure the reliability of results.

Model Building: Model building involves developing a simplified representation of a real-world system using logical and quantitative relationships. The modeller identifies the required state variables and system components according to the objectives of the study. This process usually begins with a conceptual model, followed by a mathematical or logical model, which is then implemented in a computer program for simulation experiments.

Verification: Verification is the process of checking whether the simulation program correctly **implements** the intended model. It is often described as *“building the model right.”* Debugging techniques such as using the same sequence of random numbers help identify programming errors. Animation is a valuable verification tool, as it visually confirms whether entities and events follow the specified logic of the model.

Validation: Validation determines whether the simulation model accurately **represents** the real system and its behavior, commonly referred to as “*building the right model*.” Validation is performed by comparing simulation outputs with actual system data whenever possible. Animation can also assist in validation by revealing unrealistic movements or interactions. Furthermore, the use of a statistically reliable random-number generator is essential, as valid simulation results cannot be obtained without it.

Question -2023

01.

- a) Explain the importance of system modeling in simulation studies. **(03)**
 - b) Differentiate between continuous and discrete system modeling. **(02)**
 - c) Describe the steps involved in constructing a system model. **(03)**
 - d) Discuss the advantages and limitations of using a mathematical model in system simulation. **(04)**
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a) Importance of System Modelling in Simulation Studies (03)

System modelling is crucial in simulation studies for several reasons:

- **Practicality and Safety:** Testing real systems can be expensive or risky. Simulation allows experiments without disturbing the actual system (e.g., testing staff reduction without upsetting customers).
- **Design of New Systems:** When a system does not yet exist, simulation helps compare different designs and choose the best one.
- **Handling Complexity:** Many real systems are too complex for exact mathematical solutions. Simulation can handle such complexity using numerical experiments.
- **Communication and Training:** Animation and visual models help explain the system to non-experts, assist debugging, and support staff training.
- **Reproducibility:** Using good random-number generators allows exact repetition of experiments, useful for debugging and fair comparison.

b) Difference Between Continuous and Discrete System Modelling (02)

The primary difference lies in how the state variables (the collection of variables describing the system at a particular time) change over time:

- **Discrete System Modelling:** The state variables change instantaneously at separated points in time, usually when specific events occur, such as a customer arriving or departing a bank.

- **Continuous System Modelling:** The state variables change continuously and smoothly with respect to time. An example is an airplane moving through the air, where variables like position change constantly.

c) Steps Involved in Constructing a System Model (03)

1. **Define System and Objectives:** Identify the collection of entities (e.g., people or machines) that interact toward a logical end. The model should be built to meet these specific goals.
2. **Identify State Variables and Attributes:** Determine the variables necessary to describe the system at any given time (state variables) and the data values that characterise specific entities (attributes).
3. **Establish Logical and Quantitative Relationships:** Set up the logical or mathematical rules that explain how the variables affect each other and how the system works.
4. **Define Events and Routines:** For discrete models, identify the main events and write event routines that describe the state changes occurring at those specific moments.
5. **Specify Probability Distributions:** Choose suitable probability distributions for random factors, such as arrival times or demand, so the model behaves like a real system.

d) Advantages and Limitations of Mathematical Models in System Simulation (04)

Advantages:

- **Insight through Manipulation:** They represent systems through quantitative relationships that can be easily manipulated to observe reactions, aiding in the understanding of the real system.
- **Study of Complexity:** They allow for the study of complex systems where exact mathematical solutions are impossible to find, using numerical testing instead.
- **Comparison of Policies:** They enable the comparison of alternative policies, such as different ordering strategies in an inventory system, to find the most efficient approach.

Limitations:

- **Stochastic Randomness:** Most real systems involve randomness. Because these models use random variables, their outputs are also random and provide only estimates rather than exact values.
- **Resource Intensity:** Developing and running a mathematical model for a complex, real-world system is often highly time-consuming.
- **Statistical Dependability:** The model is only as good as its underlying logic; for instance, if it lacks a statistically reliable random-number generator, it is impossible to obtain correct or valid results.

02.

a) How to develop a simulation model? (02)

b) Find out how well the current arrangement works in terms of the response time of the system. (04)

c) Write the mechanism for advancing simulation time and guaranteeing that all events occur in correct chronological order. (03)

d) Is only one snapshot (the current one or partially updated one) kept in computer memory when an event-scheduling algorithm is computerized? (03)

a) How to develop a simulation model? (02)

To develop a simulation model, the process can be explained in four simple steps:

1. **Define Objectives and System State:** First, clearly decide the purpose of the study and set the boundaries of the system. Then identify the key state variables (for example, the number of customers in a bank) that describe the system at any given time.
2. **Classify the Model:** Decide the type of simulation model to use. Most real systems are dynamic (change over time), stochastic (involve randomness), and discrete-event (changes happen at specific points in time).
3. **Organise Program Logic:** Arrange the model logic using an event-scheduling or process-oriented approach. Include important routines such as initialization, timing, and event routines to control time flow and system state changes.
4. **Add Randomness and Verify the Model:** Use proper random-number generators to model uncertain inputs like service or arrival times. Finally, test and verify the model using debugging techniques, animation, or repeated runs to ensure the model works correctly.

b) Evaluation of system response time and performance (04)

1. Measuring Average Delay

The main response-time measure is the **average delay in the queue** $\hat{d}(n)$, which is the average time a customer waits from arrival until service begins. It is calculated as:

$$\hat{d}(n) = \frac{\sum_{i=1}^n D_i}{n}$$

where D_i is the waiting time of the i -th customer.

2. Time-Average Number in Queue

System performance is also measured by the **time-average number of customers in the queue**, denoted by $\hat{q}(n)$. This shows how crowded the system is over time.

3. Validity of Results

Because arrivals and service times are **random**, one simulation run gives only an **estimate**. Managers compare these results with system goals, such as balancing service cost and customer waiting time.

c) Advancing simulation time and chronological order (03)

The timing routine controls how time moves forward in a simulation. It checks the event list to find the next event with the smallest scheduled time and advances the simulation clock directly to that time. Events are then executed one by one in the order of their occurrence to maintain correct chronological sequence. If two or more events are scheduled at the same time, a tie-breaking rule, such as giving priority to the lowest-numbered event, is applied.

d) Snapshots in computer memory (03)

Yes, in a computerized event-scheduling algorithm, only one snapshot of the system state is kept in computer memory at any time. This snapshot represents the current state of the system. When an event occurs, the system state is updated immediately, and the old state is discarded. The simulation then proceeds using this updated snapshot until the next event occurs. This approach makes the algorithm memory-efficient and practical for large simulations.

03.

- a) *"The hardest part about simulation is deciding what NOT to model."* Justify this statement. (02)
- b) Briefly describe the Monte Carlo simulation process with an example. (06)
- c) Differentiate between Continuous and Discrete Simulation Models. (04)

a) Justification of the statement:

"The hardest part about simulation is deciding what NOT to model."

In simulation, real-world systems are extremely complex and contain many interacting components. It is usually impossible and unnecessary to include every detail of the real system in a model. The main challenge for the modeler is to identify which elements are essential to the objectives of the study and which details can be safely ignored.

Including too many unnecessary details makes the model complex, slow, and difficult to understand, verify, and validate, without significantly improving the accuracy of the results. On the other hand, omitting important factors can lead to misleading or incorrect conclusions. Therefore, the modeler must carefully balance simplicity and realism, focusing only on variables and relationships that significantly influence system behavior. This decision-making process is what makes deciding *what not to model* the hardest part of simulation.

04.

- a) A computer technical support center is staffed by two people, Able and Baker, who take calls and try to answer questions and solve computer problems. The time between calls ranges from 1 to 4

minutes with the distribution shown in Table 1b(1). Able is more experienced and can provide service faster than Baker, which means that when both are idle, Able takes the call. The distribution of their service times is shown in Table 1b(2) and Table 1b(3), respectively.

Table 1b(1): Interarrival Time Distribution

Interarrival Time (minutes)	1	2	3	4
Probability	0.25	0.40	0.20	0.15

Table 1b(2): Service Time Distribution of Able

Service Time (minutes)	2	3	4	5
Probability	0.30	0.28	0.25	0.17

Table 1b(3): Service Time Distribution of Baker

Service Time (minutes)	3	4	5	6
Probability	0.35	0.25	0.20	0.20

Random digits for interarrival times are:

26, 98, 90, 26, 42, 74, 80, 68, 22, 48, 34, 45, 24, 34

Random digits for service times are:

95, 21, 51, 92, 89, 38, 13, 61, 50, 49, 39, 53, 88, 01, 81

Simulate the system for **10 customers** and find:

- i) Average interarrival time
- ii) Average service time of Baker
- iii) Average waiting time of all customers
- iv) Average waiting time of those who wait

05.

- a) What are random numbers, and why are they important in simulation studies? **(03)**
- b) Discuss the different methods for generating random numbers in simulation studies. **(03)**
- c) What are the properties of a good random number generator? **(02)**
- d) Explain how pseudo-random numbers are generated and discuss their limitations. **(04)**

a) Random Numbers and Their Importance (03)

Random numbers are values generated from a uniform distribution on (0, 1), written as $U(0,1)$. They are the basic building block of simulation, because all other random variables are generated from them. Simulation models use random numbers to represent uncertainty and randomness in real systems, such as interarrival times in a single-server queue. Without a statistically reliable random-number generator, simulation results cannot be trusted.

b) Methods for Generating Random Numbers (03)

Common methods for generating random numbers include:

1. Linear Congruential Generator (LCG)

Generates numbers using:

$$Z_i = (aZ_{i-1} + c) \bmod m, U_i = \frac{Z_i}{m}$$

It is simple and widely used.

2. Mixed and Multiplicative Generators

- **Mixed:** $c > 0$, can give full period.
- **Multiplicative:** $c = 0$, faster but shorter period.

3. Multiple Recursive Generators (MRGs)

Use several previous values to obtain longer periods:

$$Z_i = (a_1Z_{i-1} + a_2Z_{i-2} + \dots + a_qZ_{i-q}) \bmod m$$

With good parameter choices, the period can reach $m^q - 1$. MRGs are portable and statistically reliable.

4. Composite Generators

Combine generators (e.g., shuffling or combined MRGs) to improve randomness and extend period.

5. Feedback Shift Register Generators

Use feedback logic to generate long random sequences efficiently.

c) Properties of a Good Random Number Generator (02)

A good random-number generator should have:

- **Statistical quality:** Numbers should be uniformly distributed and uncorrelated.
- **Efficiency:** It should be fast and use little memory.
- **Reproducibility:** The same sequence should be generated again for debugging and comparison.
- **Portability and streams:** It should work consistently on different computers and support independent streams.

d) Generation and Limitations of Pseudo-Random Numbers (04)

Pseudo-random numbers are generated using deterministic mathematical algorithms called random-number generators (such as Linear Congruential Generators). These algorithms start with an initial value called a seed and produce a sequence of numbers that appear random. Pseudo-random numbers are generated using deterministic formulas. For example, an LCG uses

$$Z_i = (aZ_{i-1} + c) \bmod m, U_i = \frac{Z_i}{m}$$

Limitations:

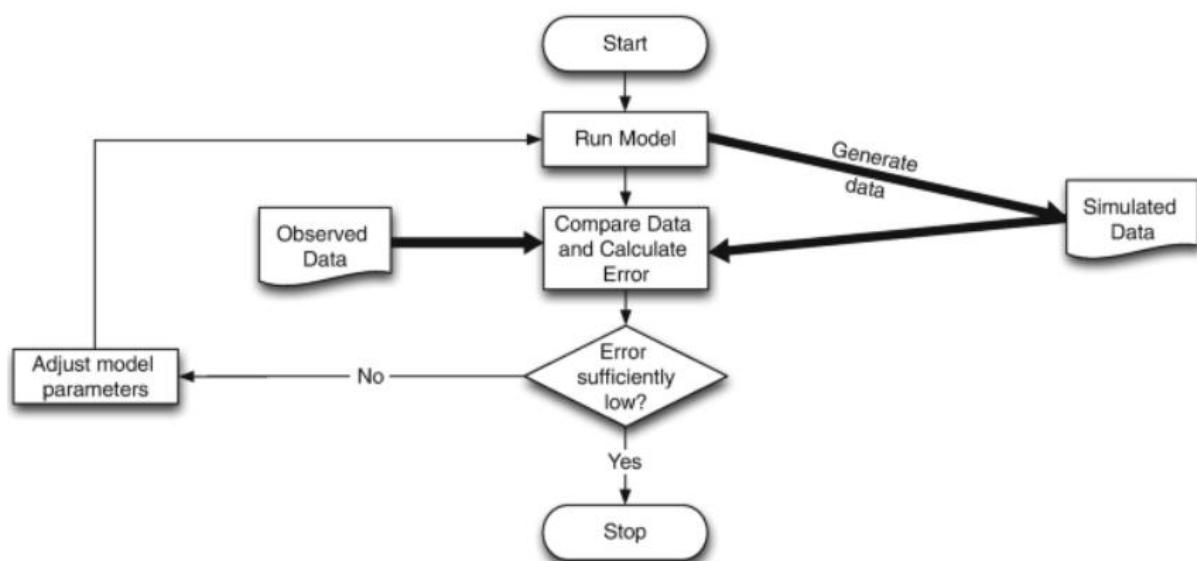
Although pseudo-random numbers appear random, they are not truly random. The sequences are periodic, meaning they eventually repeat after a fixed length. Poorly chosen generator parameters can lead to short periods, correlations, or patterns, which may reduce the reliability of simulation results. Therefore, careful design and testing of the generator are essential.

06.

- a) What is calibration? Explain it with a suitable diagram. (04)
 - b) Explain the following terms used in discrete-event simulation: (08)
 - i) List
 - ii) Event notice
 - iii) Delay
 - iv) Clock
-

a) What is Calibration? (04)

Calibration in simulation is the process of adjusting model parameters so that the simulation output matches real-world (observed) system data as closely as possible. The goal is to make sure the model behaves like the actual system before it is used for analysis or decision-making. In calibration, parameters such as arrival rates, service times, processing times, or demand distributions are tuned by comparing simulation results with real data. This process is repeated until the difference between simulated output and real system performance is acceptably small.



Calibration Diagram Description:

As shown in the diagram, the calibration process works as follows:

1. The process starts by running the simulation model using initial parameter values.
 2. The model generates simulated data as output.
 3. This simulated data is then compared with the observed (real) data, and the error is calculated.
 4. A decision is made to check whether the error is sufficiently low.
 5. If the error is not low enough, the model parameters are adjusted, and the model is run again.
 6. This cycle continues until the error becomes acceptably small.
 7. Once the error is sufficiently low, the calibration process stops.
-

b) Terms Used in Discrete-Event Simulation (08)

i) List

In simulation, a list is a data structure used to store and manage records such as events or entities. The most important list is the event list, which keeps track of all scheduled future events and their occurrence times. Lists are also used to manage queues, where entities (such as customers) wait for service. These queues can follow rules like FIFO (First-In, First-Out) or LIFO (Last-In, First-Out).

ii) Event Notice

An event notice is a record placed in the event list that represents a future event. It contains information such as the event type and the time when the event will occur. For example, when a customer arrives, the simulation creates an event notice for the next arrival and adds it to the event list.

iii) Delay

A delay is the time an entity spends waiting in the system. In queueing systems, it is defined as the time from a customer's arrival until service begins. Delays depend on random factors like arrival times and service times, so they are treated as random variables. In process-based simulation, delays are also called conditional waits.

iv) Clock

The simulation clock keeps track of the current simulated time. Unlike a real clock, it does not move continuously. Instead, the clock advances by jumping directly to the time of the next event from the event list, allowing the simulation to skip periods where no events occur.

07.

- a) Find the mean and variance of the exponential distribution. Suppose that the life of an industrial lamp, in thousands of hours, is exponentially distributed with failure rate $\lambda = \frac{1}{3}$ (one failure every 3000 hours on average). Find the probability that the lamp will last longer than its mean life. **(08)**
- b) Can you explain what modeling and simulation mean in the context of computer science? **(04)**
-

a) (a) Mean and Variance of the Exponential Distribution

Let X be an exponential random variable with **failure rate** λ .

PDF

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

Mean

$$E(X) = \frac{1}{\lambda}$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Given

- Failure rate:

$$\lambda = \frac{1}{3} \text{ (failures per 1000 hours)}$$

- Unit: **thousands of hours**

Mean life

$$E(X) = \frac{1}{\lambda} = \frac{1}{1/3} = 3$$

So, the **mean life = 3 thousand hours = 3000 hours**.

Probability that the lamp lasts longer than its mean life

We want:

$$P(X > E(X)) = P(X > 3)$$

For an exponential distribution:

$$P(X > x) = e^{-\lambda x}$$

Substitute values:

$$P(X > 3) = e^{-(1/3) \times 3} = e^{-1}$$

$$P(X > \text{mean life}) = e^{-1} \approx 0.3679$$

Question -2024(4th)

1.

- (a) Differentiate between Static and Dynamic Simulation Models.
- (b) Differentiate between Deterministic and Stochastic Simulation Models.
- (c) Differentiate between Continuous and Discrete Simulation Models.

(a) Differentiate between Static and Dynamic Simulation Models.

A static simulation model is a representation of a system at a particular point in time or a system in which time does not play any role. It focuses on analyzing the system's behavior under fixed conditions and does not consider how the system changes over time.

On the other hand, a dynamic simulation model represents a system as it evolves over time. It explicitly includes time as a key variable and shows how the state of the system changes with time. Dynamic models are used to study ongoing processes such as queues, inventory systems, or the movement of items on a conveyor system in a factory, where events occur continuously or at discrete time intervals.

(b) Differentiate between Deterministic and Stochastic Simulation Models.

A simulation model is called deterministic if it does not include any probabilistic (random) components. In a deterministic model, once the input values and the relationships between them are defined, the output is fixed and always the same.

In contrast, many real systems involve some degree of randomness in their inputs. These systems are represented using stochastic simulation models. Most queueing systems are modeled stochastically. Since stochastic models include random variables, their outputs are also random and cannot be exact. Therefore, the results obtained from stochastic simulation models are treated only as estimates of the true behavior of the system.

(c) Differentiate between Continuous and Discrete Simulation Models.

A continuous simulation model represents a system in which the state variables change continuously over time. The changes happen smoothly, not at specific moments.

A discrete simulation model, on the other hand, represents a system in which changes in the system state occur only at specific points in time, usually when certain events happen, such as arrivals or departures in a queueing system.

It should be mentioned that a discrete model is not always used to model a discrete system, and a continuous model is not always used to model a continuous system. The choice of model depends on

how the system behavior is best represented for the purpose of the study, not only on the nature of the real system.

2.

(a) What are the key differences between event-scheduling approach and process-interaction approach? Which approach can be considered better?

(b) Can Continuous and Discrete Simulation Models be separated in real-world systems? Explain with examples.

(b) Can Continuous and Discrete Simulation Models be separated in real-world systems? Explain with examples.

In real-world systems, continuous and discrete simulation models are not completely separate, because most practical systems are not purely one type. Although models are classified based on how their state variables change, the choice of a simulation model usually depends on the goal of the study.

Conceptual Differences

Discrete Simulation Models: A discrete simulation model, on the other hand, represents a system in which changes in the system state occur only at specific points in time, usually when certain events happen, such as arrivals or departures in a queueing system.

Continuous Simulation Models: A continuous simulation model represents a system in which the state variables change continuously over time. The changes happen smoothly, not at specific moments.

Interconnectedness in Practice

In practice, a discrete model is not always used only for a discrete system, and a continuous model is not always used only for a continuous system. Most real-world systems show features of both types:

- **Example of a Discrete System (Bank):**
A bank is a typical discrete system. State variables such as the number of customers in the bank or the number of busy tellers change only when specific events occur, like customer arrivals or departures.
- **Example of a Continuous System (Airplane):**
An airplane moving through the air is a continuous system because its state variables, such as position and speed, change continuously with time.
- **Example of Hybrid Characteristics (Inventory System):**
In an inventory simulation, the number of items demanded is a discrete random variable because it involves whole numbers (1, 2, 3, etc.). However, the time between demands is often modeled using a continuous probability distribution, such as the exponential distribution.

Furthermore, even in discrete-event simulations, some performance measures are calculated over continuous time (for example, the average number of customers in a queue over time), while others are calculated over discrete units (such as the average waiting time per customer). This shows that discrete and continuous concepts are often combined when modeling real-world systems.

3.

(a) What are the advantages and disadvantages of simulation?

(b) Briefly discuss the Monte Carlo simulation process with a suitable example.

(a) What are the advantages and disadvantages of simulation?

Advantages of Simulation

- **Practicality and Safety:** Testing real systems can be expensive or risky. Simulation allows experiments without disturbing the actual system (e.g., testing staff reduction without upsetting customers).
- **Design of New Systems:** When a system does not yet exist, simulation helps compare different designs and choose the best one.
- **Handling Complexity:** Many real systems are too complex for exact mathematical solutions. Simulation can handle such complexity using numerical experiments.
- **Communication and Training:** Animation and visual models help explain the system to non-experts, assist debugging, and support staff training.
- **Reproducibility:** Using good random-number generators allows exact repetition of experiments, useful for debugging and fair comparison.

Disadvantages of Simulation

- **Randomness and Inaccuracy:** Results are estimates because outputs depend on random variables, not exact values.
- **Resource Intensity:** Simulations can be time-consuming and costly to develop and run.
- **Statistical Risks:** Poor random-number generators can lead to incorrect and unreliable results.
- **Software Limitations:** Some simulation tools lack flexibility and may run slower than custom programs written in languages like C++ or Java.

b) Monte Carlo Simulation Process

1. **Define Random Components:** The basic requirement of Monte Carlo simulation is the generation of random numbers from a uniform distribution $U(0,1)$. These numbers are independent and lie between 0 and 1.
2. **Generate Random Variates:** The $U(0,1)$ random numbers are transformed into random variates that follow required probability distributions to represent uncertainty in the system.
3. **Numerical Experimentation:** Since the model contains random elements, its outputs are also random. The simulation is run many times, and the results are treated as estimates of the real system's behaviour. In Monte Carlo simulation, time does not play a role.

4. **Collect and Analyze Results:** Performance measures are stored using statistical counters, and final estimates are obtained by averaging the results over many simulation runs.

Example: Estimation of π Using Monte Carlo Simulation

- **Idea:**
The value of π can be estimated using random points inside a **unit square**.

Procedure

1. Generate random pairs (x, y) , where $x, y \sim U(0,1)$.
2. Count how many points fall **inside the quarter circle** defined by:

$$x^2 + y^2 \leq 1$$

3. Let:
 - N = total number of random points
 - M = number of points inside the quarter circle
4. Estimate π as:

$$\pi \approx 4 \times \frac{M}{N}$$

As the number of random points increases, the estimate of π becomes more accurate. This method avoids complex mathematical formulas and relies only on random sampling.

4.

- (a) Describe the simulation of a Single-Server Queueing System.
- (b) Describe the simulation of an Inventory System with a suitable **program organization and logic**.

b) The simulation of an inventory system is a discrete-event study used to compare ordering policies over a fixed period of n months. In this system, state variables such as the inventory level change instantaneously whenever an event occurs.

1. System Logic and Policy

The system follows a stationary (s, S) policy. At the beginning of each month, the inventory level I is reviewed:

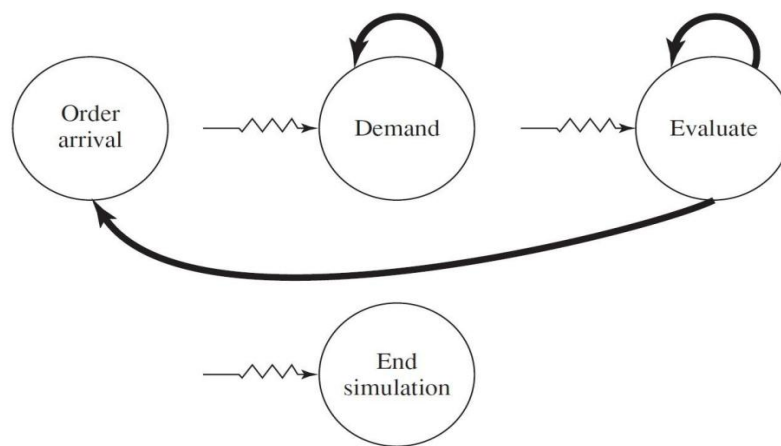
- If $I < s$, an order of size $Z = S - I$ is placed.
- If $I \geq s$, no order is placed.

The system includes a fixed ordering cost K , a per-item cost i , holding costs for positive inventory I^+ , and shortage costs for negative inventory I^- .

2. Program Organisation

The simulation is organised around four event types, processed in time order:

1. **Order Arrival:** Inventory increases when an order arrives after a random lead time.
2. **Demand:** Random customer demand reduces inventory and schedules the next demand.
3. **Inventory Evaluation:** Occurs at the start of each month to apply the (s, S) policy.
4. **End of Simulation:** Stops the simulation at month n .



3. Handling Simultaneous Events

When multiple events occur at the same time, the timing routine gives priority to the lowest-numbered event. For example, at month n , the end-of-simulation event is processed before inventory evaluation to avoid unnecessary ordering costs.

4. Randomness and Statistical Collection

Random values are generated using library routines:

- Inter-demand times follow an exponential distribution.
- Demand sizes are generated from $U(0,1)$ using cumulative probabilities.
- Lead times are uniformly distributed between 0.5 and 1 month.

A non-event routine updates the time-average inventory and backlog by calculating the area under $I^+(t)$ and $I^-(t)$ after each event.

5.

(a) In a Single-Server Queueing System, arrivals occur at times

0.4, 1.0, 2.1, 3.8, 4.0, 5.6, 5.8, 7.2

Departures (service completions) occur at times

2.4, 3.1, 3.3, 5.6, and 8.6.

Find the **simulation end time (T_n)** and **Gantt chart ($y(t)$)**.

(b) A service facility consists of **two servers in series (tandem)**, each with its own FIFO queue.

- Customers completing service at server 1 proceed to server 2,
- Customers completing service at server 2 leave the facility.
- Interarrival times are IID exponential with mean **9 minutes**.
- Service times at server 1 are IID exponential with mean **0.7 minutes**,
- Service times at server 2 are IID exponential with mean **0.9 minutes**.

For **T = 90 minutes**, estimate for **each server**:

1. Expected **average delay in queue**,
2. Expected **average number of customers in queue**,
3. Expected **utilization**.

b) Given: Tandem (Series) Queue System

Two servers in series, each with its own FIFO queue.

Flow:

Arrivals → Queue 1 → Server 1 → Queue 2 → Server 2 → Departure

Step 1: Convert given means to rates

Arrival process

- Mean interarrival time = **9 minutes**

$$\lambda = \frac{1}{9} \approx 0.111 \text{ customers/min}$$

Server 1

- Mean service time = **0.7 minutes**

$$\mu_1 = \frac{1}{0.7} \approx 1.429 \text{ customers/min}$$

Server 2

- Mean service time = **0.9 minutes**

$$\mu_2 = \frac{1}{0.9} \approx 1.111 \text{ customers/min}$$

Step 2: Arrival rates to each server

Since this is a **tandem Jackson network**:

$$r_1 = \lambda = 0.111, r_2 = r_1 = 0.111$$

Step 3: Utilization of each server

Server 1

$$\rho_1 = \frac{r_1}{\mu_1} = \frac{0.111}{1.429} \approx 0.078$$

Server 2

$$\rho_2 = \frac{r_2}{\mu_2} = \frac{0.111}{1.111} = 0.10$$

✓ Both servers are **stable** ($\rho < 1$).

Step 4: M/M/1 Performance Formulas

For an M/M/1 queue:

$$L_q = \frac{\rho^2}{1 - \rho}, W_q = \frac{L_q}{r}, \text{Utilization} = \rho$$

Server 1 Calculations

Average number in queue

$$L_{q1} = \frac{(0.078)^2}{1 - 0.078} = \frac{0.0061}{0.922} \approx 0.0066 \text{ customers}$$

Average delay in queue

$$W_{q1} = \frac{L_{q1}}{r_1} = \frac{0.0066}{0.111} \approx 0.059 \text{ minutes}$$

Utilization

$$\boxed{\rho_1 \approx 0.078 \text{ (7.8\%)}}$$

Server 2 Calculations

Average number in queue

$$L_{q2} = \frac{(0.10)^2}{1 - 0.10} = \frac{0.01}{0.90} \approx 0.0111 \text{ customers}$$

Average delay in queue

$$W_{q2} = \frac{L_{q2}}{r_2} = \frac{0.0111}{0.111} = 0.10 \text{ minutes}$$

Utilization

$$\boxed{\rho_2 = 0.10 \text{ (10\%)}}$$

Step 5: Final Results (Exam-Ready Table)

Server	Avg. delay in queue W_q	Avg. number in queue L_q	Utilization ρ
1	0.059 min	0.0066 customers	0.078 (7.8%)
2	0.10 min	0.0111 customers	0.10 (10%)

Important note about $T = 90$ minutes

Since both queues are stable, these values represent the expected long-run averages, which are good estimates for a 90-minute observation period.

6.

- (a) Prove that the variance of an exponential distribution is $1/\lambda^2$.
- (b) Explain how the Inverse Transform Method is used to generate random variates from an exponential distribution.

(a) Proof of the Variance of an Exponential Distribution

The sources define the exponential distribution using the parameter β , where $\beta = 1/\lambda$ in standard notation. We show that the variance of an exponential random variable is β^2 , or equivalently $1/\lambda^2$.

Step 1: Probability Density Function (PDF)

For an exponential distribution, the probability density function is:

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, x \geq 0$$

and $f(x) = 0$ otherwise.

Step 2: Expected Value (Mean)

The expected value is defined as:

$$E(X) = \int_0^{\infty} x f(x) dx = \int_0^{\infty} x \frac{1}{\beta} e^{-x/\beta} dx$$

Using **integration by parts**, this evaluates to:

$$E(X) = \beta$$

Step 3: Variance Formula

The variance of a random variable is given by:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

To compute $E(X^2)$, we evaluate:

$$E(X^2) = \int_0^{\infty} x^2 \frac{1}{\beta} e^{-x/\beta} dx$$

Applying integration by parts twice gives:

$$E(X^2) = 2\beta^2$$

Substituting into the variance formula:

$$\text{Var}(X) = 2\beta^2 - \beta^2 = \beta^2$$

Step 4: Relation to λ

Since $\lambda = 1/\beta$, the variance can be written as:

$$\text{Var}(X) = \beta^2 = \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

(b) Generating Random Variates Using the Inverse Transform Method

The Inverse Transform Method is used to generate random variates from a specified probability distribution using a uniform random number $U \sim U(0,1)$.

Exponential Distribution (Inverse Transform Method)

Let X be a random variable that follows an exponential distribution with mean β .

Cumulative Distribution Function (CDF)

$$F(x) = \begin{cases} 1 - e^{-x/\beta}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

For $x \geq 0$,

$$F(x) = 1 - e^{-x/\beta}$$

Inverse Transform Method

Let

$$U \sim U(0,1)$$

Set the CDF equal to U :

$$U = 1 - e^{-x/\beta}$$

Rearranging,

$$e^{-x/\beta} = 1 - U$$

Taking natural logarithm on both sides,

$$-\frac{x}{\beta} = \ln(1 - U)$$

Multiplying both sides by $-\beta$,

$$x = -\beta \ln(1 - U)$$

Since both U and $1 - U$ are uniformly distributed on $(0, 1)$, this simplifies to:

$$X = -\beta \ln U$$

Algorithm

1. Generate a random number $U \sim U(0,1)$.
2. Compute

$$X = -\beta \ln U$$

The value X follows an **exponential distribution** with mean β .

7.

- (a) Describe Feedback Shift Register Generators.
- (b) Explain Naylor and Finger Simulation Model Validation Process.
- (c) Explain Transient and Steady-State behavior of a Stochastic Process.

(a) Feedback Shift Register Generators

Feedback generators are a type of pseudo-random number generator (PRNG) in which previously generated numbers are fed back into the formula to produce the next number. General form

A typical feedback generator can be written as:

$$X_{n+1} = f(X_n, X_{n-1}, \dots)$$

where earlier values are used again (feedback).

8. (a) Consider a loading dock that has two tracks and no place for trucks to wait.

- If a truck arrives when the dock is full, it must leave.
- All other arriving trucks go to other loading docks.
- When a train arrives, it enters the turned-away track and begins unloading immediately.
- Truck arrivals follow a **Poisson process** with mean arrival rate $\lambda = 2$ trucks per hour.
- Loading/unloading time is exponentially distributed with mean $\mu^{-1} = 120$ minutes (service rate $\mu = 0.5$ per hour).

Since there is a very large population of potential arriving trucks, the system is modeled as an **M/M/1/1 queue**.

Suppose at time 0 the dock is empty.

- Interarrival times (in minutes):
 $A_1 = 10, A_2 = 25, A_3 = 45, A_4 = 15, A_5 = 20$
- Service times (in minutes):
 $S_1 = 35, S_2 = 20, S_3 = 60, S_4 = 15, S_5 = 14$

Tasks:

1. Draw the **history of the loading dock**,
2. Find the **server utilization** for the system.

Ans: Given

- System type: **M/M/1/1** (one server, **no waiting space**)
- Truck arrival rate:
$$\lambda = 2 \text{ trucks/hour}$$
- Service rate:
Mean service time = 120 minutes $\Rightarrow$
$$\mu = 0.5 \text{ per hour}$$
- Initial condition: **dock empty at time 0**

Given sample data

Interarrival times (minutes)

$$A_1 = 10, A_2 = 25, A_3 = 45, A_4 = 15, A_5 = 20$$

Service times (minutes)

$$S_1 = 35, S_2 = 20, S_3 = 60, S_4 = 15, S_5 = 14$$

Task 1: Draw the history of the loading dock

Step 1: Arrival times (cumulative)

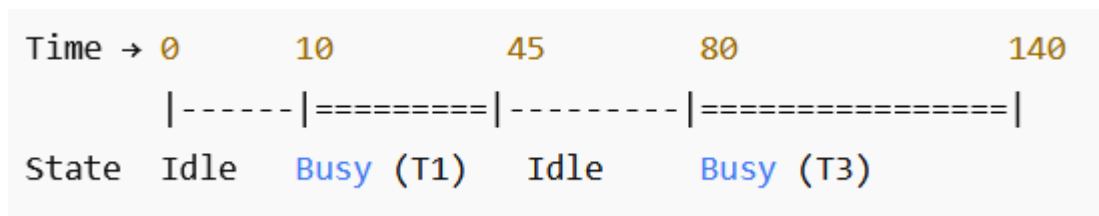
Truck	Interarrival	Arrival time
1	10	10
2	25	35
3	45	80
4	15	95
5	20	115

Step 2: Event-by-event simulation

(Remember: **if the dock is busy, the truck leaves**)

Truck	Arrival	Service Time	Service Start	Service End	Accepted / Rejected
1	10	35	10	45	Accepted
2	35	20	—	—	Rejected (dock busy)
3	80	60	80	140	Accepted
4	95	15	—	—	Rejected (dock busy)
5	115	14	—	—	Rejected (dock busy)

History of the loading dock (timeline)



Busy intervals:

- 10 → 45
- 80 → 140

Task 2: Server Utilization

Definition

$$\text{Utilization} = \frac{\text{Total busy time}}{\text{Total simulation time}}$$

Total busy time

$$(45 - 10) + (140 - 80) = 35 + 60 = 95 \text{ minutes}$$

Total simulation time

$$T = 140 \text{ minutes}$$

Utilization

$$\text{Utilization} = \frac{95}{140} = 0.6786$$

$$\boxed{\text{Server utilization} \approx 67.9\%}$$

Final Answers

1. History of the loading dock

- Truck 1 served from **10–45**
- Truck 3 served from **80–140**
- Trucks 2, 4, and 5 were **turned away**

2. Server utilization

Server utilization $\approx 67.9\%$

Question-2023

1.

- a) Define a system. Write the components of a system with an example. (4)
- b) Explain the characteristics of an open system. (4)
- c) Define the following terms with examples: (2+2)
 - i) Open system
 - ii) Closed system

a) Definition and Components of a System (4 Marks)

Definition

A system is a group of entities, such as people or machines, that work together and interact with one another to achieve a specific goal.

Components and Example

The main components of a system are:

- **Entities:** The objects that move through the system, such as **customers** or **products**.
- **Attributes:** The properties or data related to each entity, for example a **customer's arrival time**.
- **Resources:** Items or services that entities need to complete a task, such as a **machine** or a **server**.
- **Queues:** Waiting lines where entities stay when resources are busy.
- **System State:** A set of variables that describe the condition of the system at a particular time, such as the **number of busy workers**.

Example:

In a bank system studied to decide how many staff are needed, the customers are the entities and the tellers are the resources. The system state includes how many customers are waiting in the queue and whether each teller is busy or free.

b) Characteristics of an Open System

An **open system** has the following key characteristics:

1. **Interaction with Environment:** An open system continuously interacts with its external environment by exchanging information, energy, or materials.
2. **Input and Output Mechanism:** It receives inputs from the environment, processes them, and produces outputs that are sent back to the environment.
3. **Dynamic Behavior:** The behavior of an open system changes over time due to environmental influence and internal interactions.
4. **Feedback and Adaptability:** Open systems use feedback from the environment to adjust and improve their performance.

c) Define the following terms with examples (2 + 2)

i) Open System: An open system is a system that interacts with its environment by exchanging information, energy, or materials.

Example: A business organization is an open system because it takes inputs such as raw materials and labor, processes them, and produces goods or services while continuously interacting with customers, competitors, and government regulations.

ii) Closed System: A closed system is a system that does not interact, or interacts very little, with its external environment.

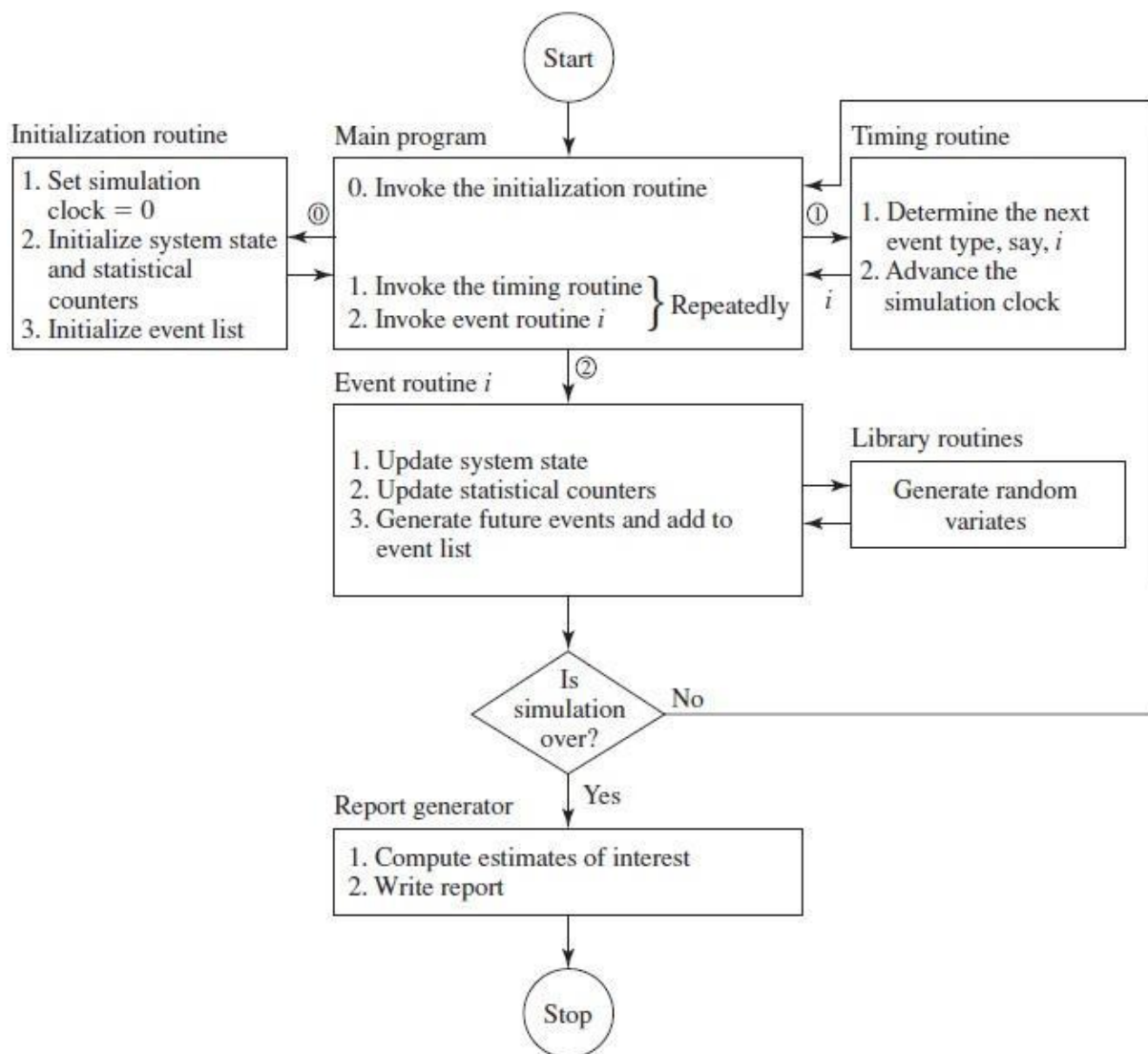
Example: A sealed chemical reaction in a laboratory is a closed system because no matter enters or leaves the container, even though heat may be exchanged.

2. a) What are the significance of modeling a system? (5)

b) Define a model. Differentiate between static and dynamic models. (5)

c) Draw a flowchart that shows the steps in a simulation study. (5)

c) Draw a flowchart that shows the steps in a simulation study.



3.

a) Write exponential growth models and exponential decay models for system dynamics with an example. (6)

b) Briefly explain the commonly required statistics for system simulation. (4)

c) What do you mean by discrete system simulation? (2)

a) Exponential Growth and Exponential Decay Models in System Dynamics (6 Marks)

In **system dynamics and simulation**, exponential models are widely used to represent random events over time.

1) Exponential Decay Model

The **exponential decay model** describes situations where the **probability of an event occurring decreases as time increases**. Shorter time intervals are more likely than longer ones.

- **Mathematical model (PDF):**

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, x \geq 0$$

- **Parameter:**
 β is the **mean time between events**.
- **Example:**
In a **queueing system**, the time between consecutive customer arrivals is often modeled using an exponential decay distribution, where customers are more likely to arrive soon than after a long delay.

2) Exponential Growth Model

The **exponential growth model** represents the **increasing probability that an event has occurred by time x** .

- **Mathematical model (CDF):**

$$F(x) = 1 - e^{-x/\beta}, x \geq 0$$

- **System logic:**
As time increases, $F(x)$ grows and approaches 1, meaning the event is almost certain to occur eventually.
- **Example:**
In a **service facility** such as a **barbershop or airport help desk**, the probability that at least one customer arrives within the next x minutes increases according to this exponential growth model.

b) Briefly explain the commonly required statistics for system simulation

In system simulation, statistics are collected to **evaluate system performance** and to **compare alternative system designs or policies**. The commonly required statistics include:

1. **Time-Average Measures**
These describe the average behavior of the system over time, such as average queue length, average inventory level, or average number of customers in the system.
2. **Event-Based Measures**
These are calculated at specific events, for example waiting time of customers, service time, or time between arrivals.
3. **Probability Measures**
These give the likelihood of certain conditions, such as the probability of system congestion, probability of stockout, or probability that a server is idle.
4. **Cost and Utilization Measures**
These include total cost, average operating cost, and utilization of resources (such as server utilization or machine usage).

Since simulation systems often involve randomness, the results are considered estimates, not exact values.

7.

- a) What is a random number? List down the characteristics of a good random number. (2+2)
 - b) Explain LCG. What are the conditions to find full cycle random numbers from this algorithm? (5)
 - c) Develop an algorithm to generate 10 random numbers. (3)
-

(b) Linear Congruential Generators (LCG)

A Linear Congruential Generator (LCG) is a commonly used arithmetic technique for generating a sequence of random numbers that appear to be independent and uniformly distributed over a fixed interval. The generator produces a sequence of integers $Z_1, Z_2, \dots$ using the recursive relation:

$$Z_i = (aZ_{i-1} + c) \bmod m$$

The corresponding random numbers U_i are obtained by normalizing the integers with respect to the modulus:

$$U_i = \frac{Z_i}{m}$$

The parameters of the algorithm are defined as follows:

- **m (Modulus):** A large positive integer that determines the range of generated values.
- **a (Multiplier):** A constant such that $0 < a < m$.
- **c (Increment):** A constant such that $0 \leq c < m$.
- **Z_0 (Seed):** The initial value of the sequence, where $0 \leq Z_0 < m$.

Conditions for Full Period (Hull–Dobell Theorem)

An LCG has a **full period** if:

1. $\gcd(m, c) = 1$
2. Every prime factor of m divides $(a-1)$
3. If m is divisible by 4, then $(a-1)$ is also divisible by 4

Note: A multiplicative LCG (where $c = 0$) can never achieve a full period because it does not satisfy the first condition.

(c) Algorithm to Generate 10 Random Numbers

Based on the principles of the **Linear Congruential Generator**, the following algorithm can be used to generate **10 random numbers**:

1. **Initialization:**

- Select appropriate values for the modulus m , multiplier a , and increment c that meet statistical requirements.
- Choose an initial seed Z_0 to begin the sequence.

2. Generation Loop:

- For $i = 1$ to 10, perform the following steps:
 - Compute the next integer value using

$$Z_i = (aZ_{i-1} + c) \bmod m$$
 - Convert it into a random number using

$$U_i = \frac{Z_i}{m}$$
 - Output or store the value of U_i .
 - Update the previous value by setting $Z_{i-1} = Z_i$.

3. Termination:

- Stop the process after generating 10 values of U_i .

8.

- What is continuous simulation? Discuss with an example. (4)
- Define calibration. Explain it with necessary diagram. (5)
- What is physical time? Differentiate between physical and simulation time. (3)

a) What is continuous simulation? Discuss with an example.

A continuous simulation model is one used to represent a system in which the state variables change continuously with respect to time. Unlike discrete models, where changes occur instantaneously at specific, separated points in time (events), the changes in a continuous model happen smoothly and constantly over the duration of the simulation.

Example: Airplane in Flight

A primary example of a continuous system is an airplane moving through the air. In this scenario:

- The state variables (such as the airplane's geographical position, altitude, or velocity) change continuously as time progresses.

c) What is Physical Time? Differentiate between Physical Time and Simulation Time

Physical Time:

Physical time is the **actual, real-world time** measured by clocks and calendars. It represents the true passage of time in a real system (seconds, minutes, hours, days).

Difference between Physical Time and Simulation Time

Aspect	Physical Time	Simulation Time
Definition	Real-world time as it actually occurs	Artificial time used inside a simulation model
Measurement	Measured by clocks and calendars	Measured by a simulation clock
Speed	Advances at a fixed real rate	Can advance faster, slower, or in jumps
Usage	Exists in real systems	Used to study and analyze system behavior

